INCH-POUND
MIL-DTL-24231/22E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/22D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, TYPE IV, 80-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 6.

2. Each nut, washer, and receptacle body shall be examined for cracks in accordance with the liquid penetrant method of T9074-AS-GIB-010/271. Finding of cracks shall be cause for rejection.

3. Welding of the pin is allowed after the hull fitting body has been age-hardened to achieve its required physical properties.

4. All dimensions are in inches.

5. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

6. These connectors are for maintenance purposes only.

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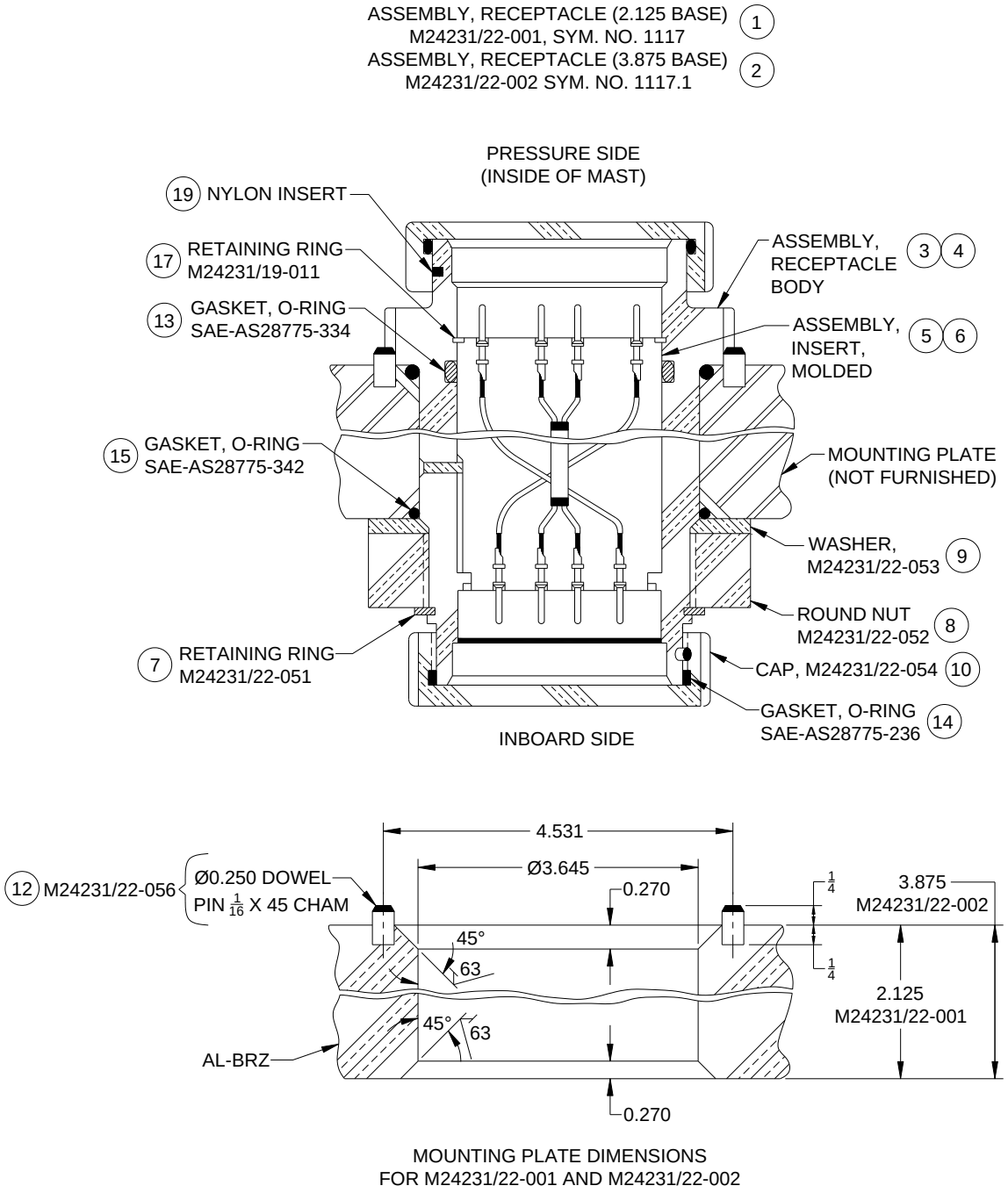
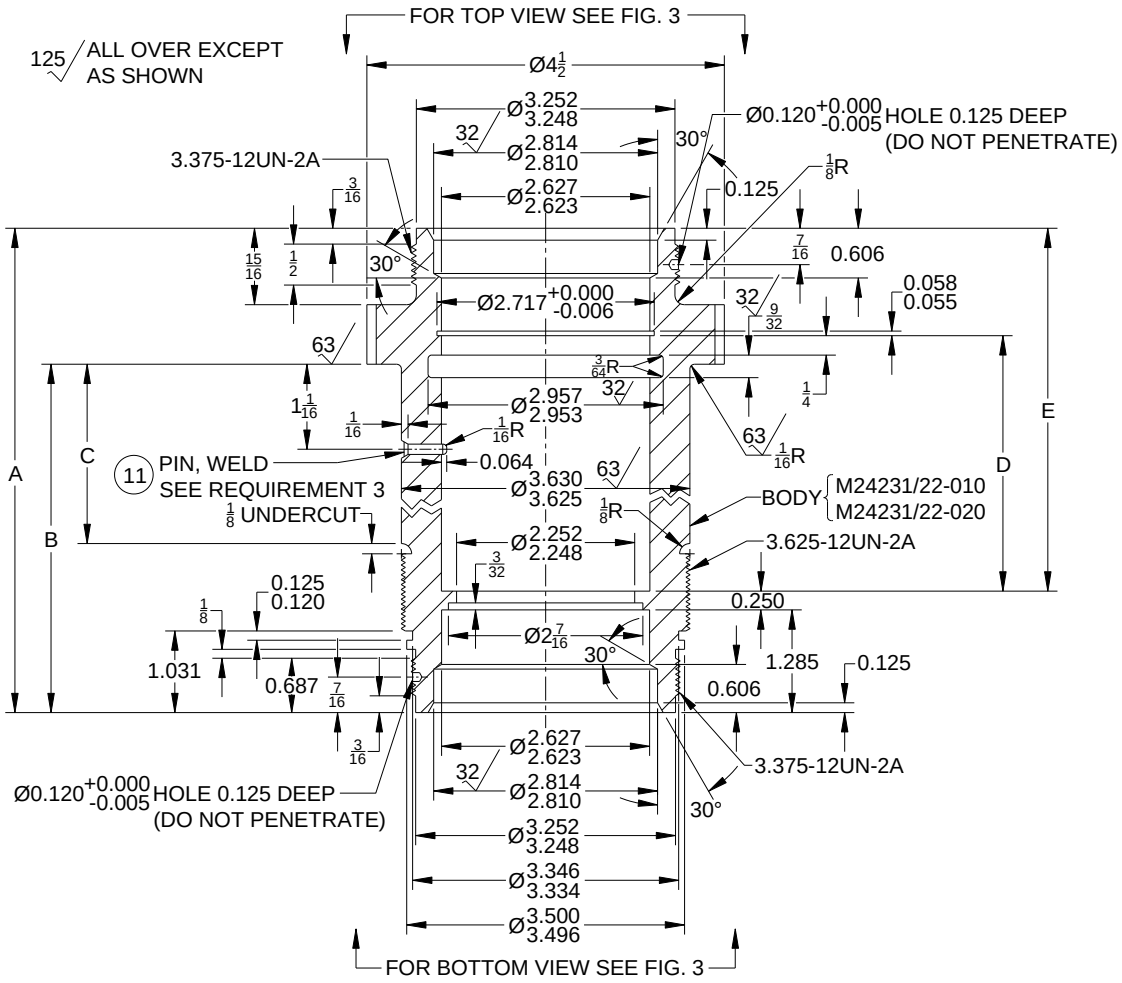


FIGURE 1. Connector, receptacle, 80-conductor – typical.

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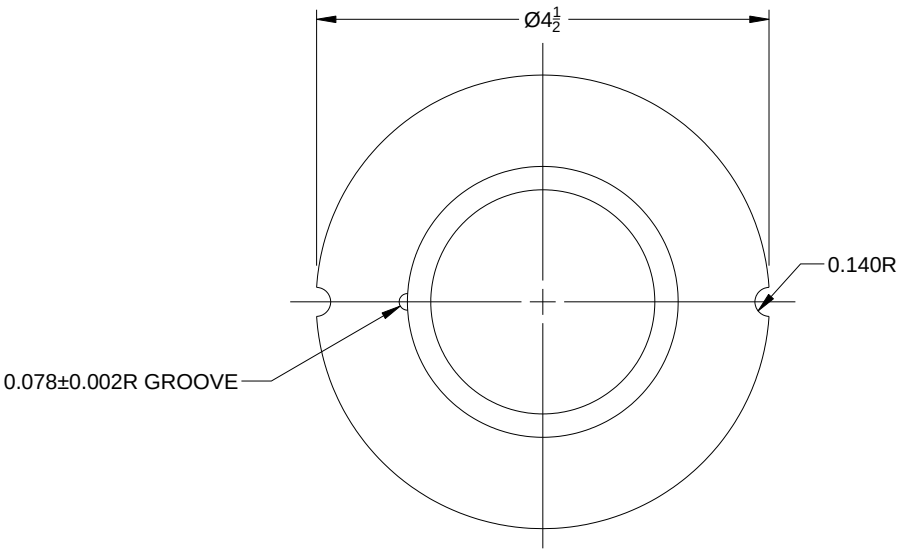


ASSEMBLY
RECEPTACLE BODY { M24231/22-010 M24231/22-020

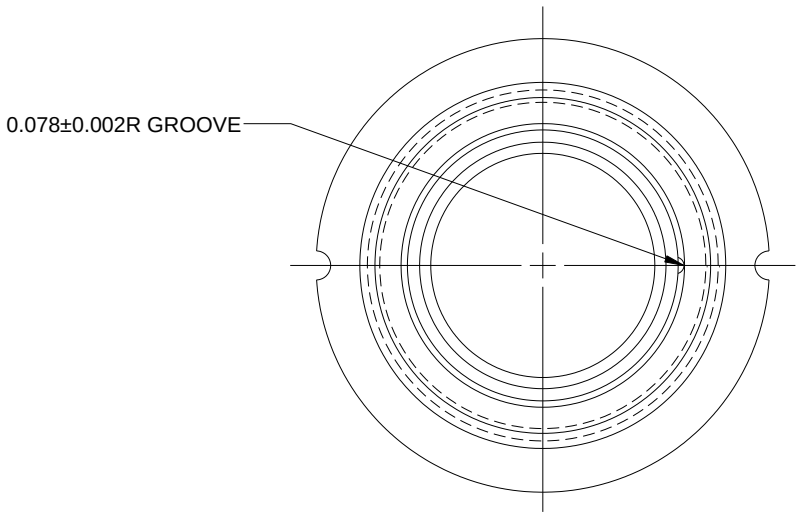
TABLE OF DIMENSIONS FOR M24231/22-010 M24231/22-020		
DIM	M-22-010	M-22-020
A	6.062	7.812
B	4.375	6.125
C	2.250	4.000
D	3.204 3.194	4.954 4.944
E	4.547 4.537	6.297 6.287

FIGURE 2. Assembly, receptacle body. (3)(4)

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TOP VIEW



BOTTOM VIEW

FOR SECTIONAL VIEW SEE FIGURE 2

FIGURE 2. Assembly, receptacle body – Continued. (3)(4)

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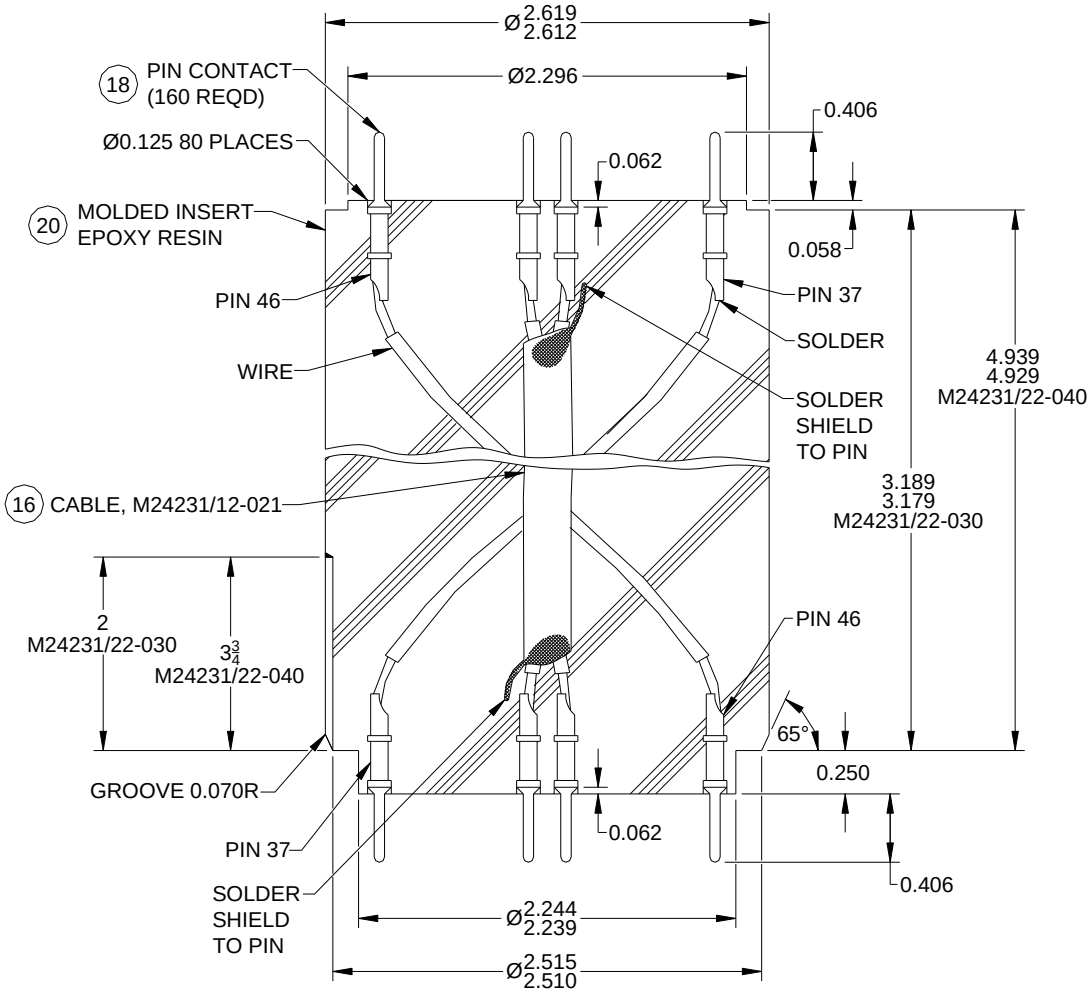


FIGURE 3. Assembly, insert, molded, M24231/22-030, M24231/22-040. (5)(6)

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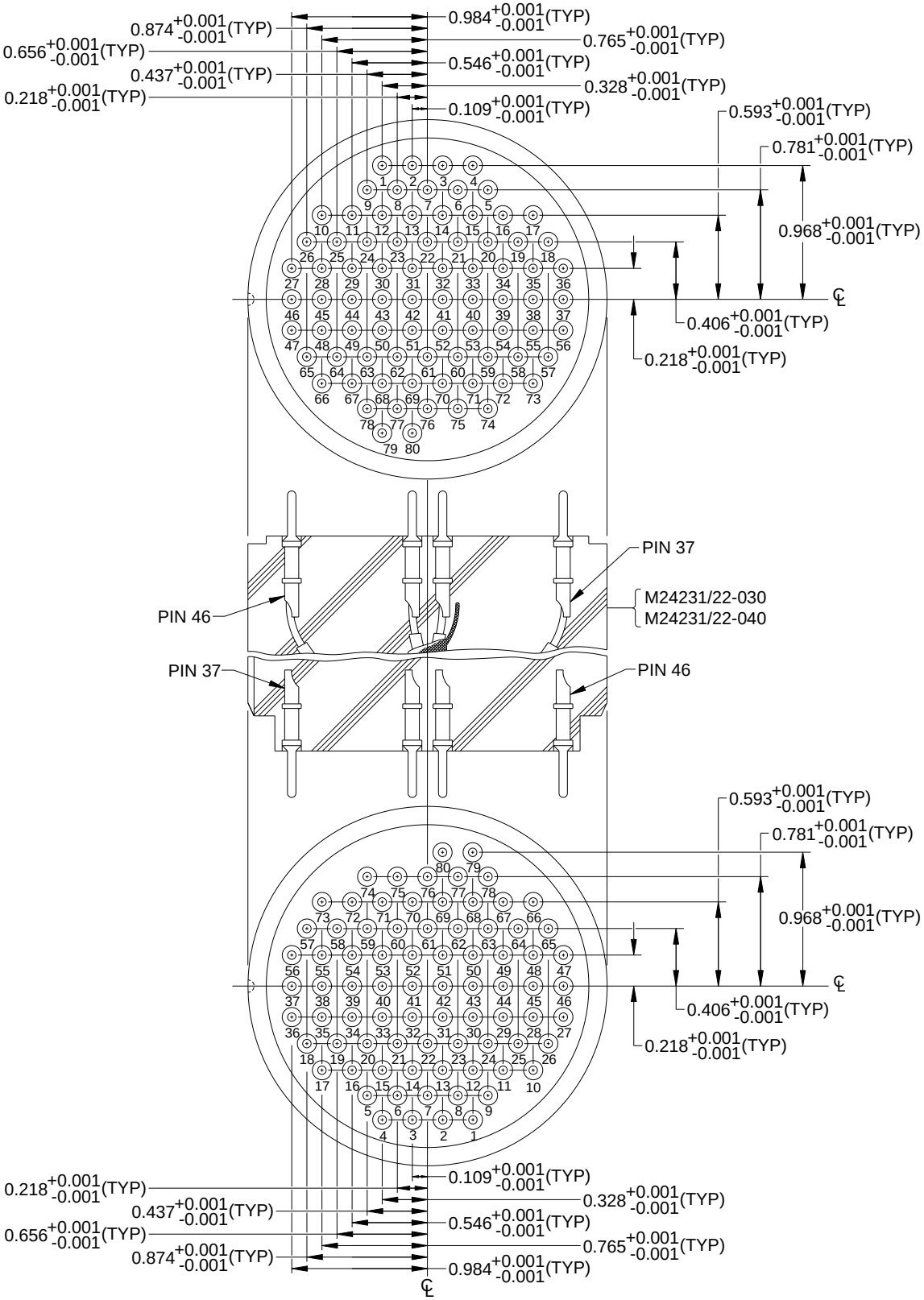


FIGURE 4. Insert, pin contact layout.

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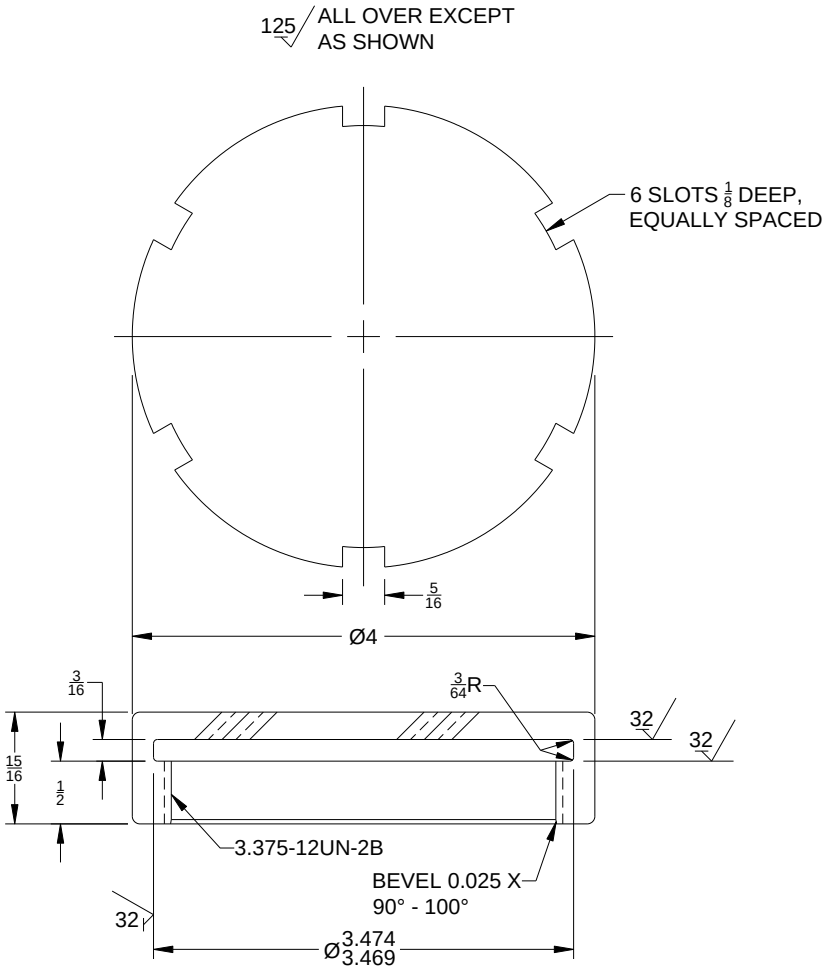


FIGURE 5. Cap – M24231/22-054. (10)

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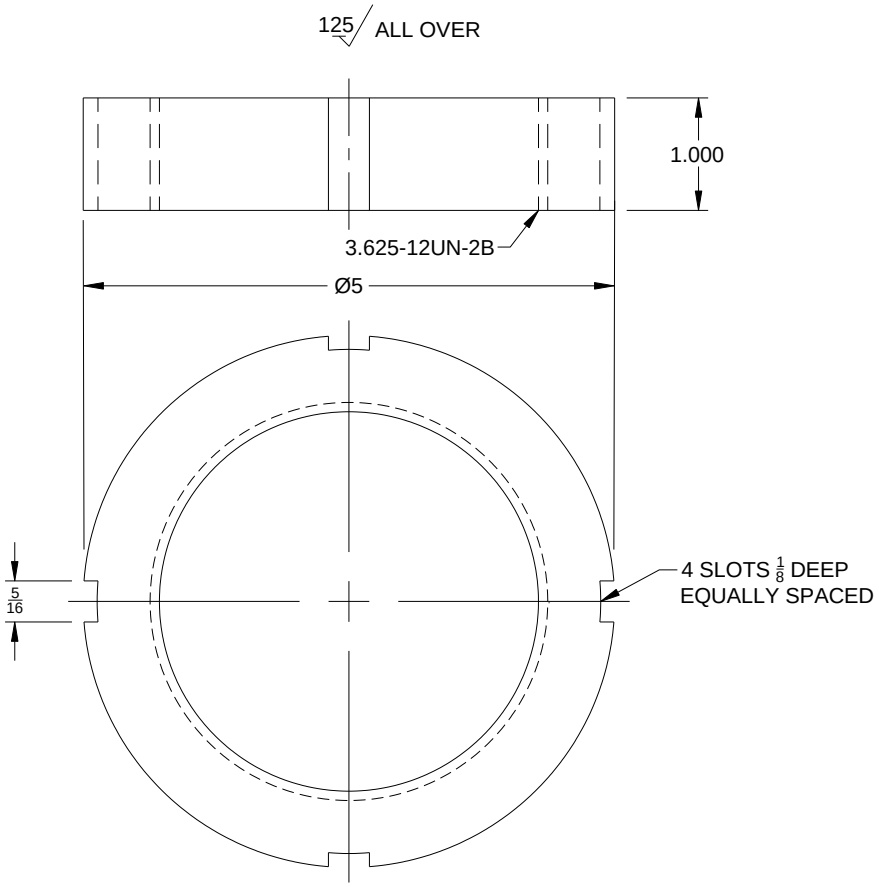


FIGURE 6. Nut – M24231/22-052.

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/22- 3/		Material/specification	Remarks
		001	002		
1	Receptacle, 80-conductor, symbol no. 1117	1		-----	See figure 1.
2	Receptacle, 80-conductor, symbol no. 1117.1		1	-----	See figure 1.
3	Body, receptacle M24231/22-010	1		Ni-Cu-Al, QQ-N-286	See figure 2.
4	Body, receptacle M24231/22-020		1	Ni-Cu-Al, QQ-N-286	See figure 2.
5	Insert assembly, molded	1		See items 16, 18, and 20.	See figure 3.
6	Insert assembly, molded		1	See items 16, 18, and 20.	See figure 3.
7	Ring, retaining	1	1	SPIROLOX RST-350-S 4/	See figure 1.
8	Nut, round	1	1	Ni-Cu-Al, QQ-N-286	See figure 1.
9	Washer, 3.640/3.635 by 5.187 by 0.187	1	1	Ni-Cu-Al, QQ-N-286	See figure 1.
10	Cap, pressure-proof	2	2	Ni-Al-BRZ, ASTM B150/B150M 5/	See figure 5.
11	Pin, guide, 0.125 diameter, length as required	1	1	Ni-Cu, Class A, QQ-N-281	See figure 2.
12	Pin, dowel	2	2	Ni-Cu, Class A, QQ-N-281	See figure 1.
13	Gasket, O-ring	1	1	SAE-AS28775-334 6/	Lubricate with silicone compound, SAE-AS8660, see figure 1.
14	Gasket, O-ring	2	2	SAE-AS28775-236 6/	See item 13.
15	Gasket, O-ring	2	2	SAE-AS28775-342 6/	See item 13.
16	Cable, 2-conductor, shielded, LS2SWU-1, length as required	8	8	MIL-DTL-24643/33	See figure 3.
17	Ring, retaining	1	1	M24231/19-011	See figure 1.
18	Pin, contact	160	160	MIL-DTL-24231	See figure 3.
19	Insert, 0.120 diameter, length as required	2	2	Nylon, Group 1, Class 1, ASTM D4066	See figure 1.
20	Epoxy resin, molded	As required		Cast epoxy resin compound 7/	See figure 3.

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TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Manufactured by Ramsey Corp., St. Louis, MO, or equal.

5/ May be high impact plastic, PVC, coml at the option of the manufacturer.

6/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

7/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-071)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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